MIL-S-11039E
2 August 1978
SUPERSEDING
MIL-S-11039D
1 November 1967

MILITARY SPECIFICATION

SLOTTED TEMPLATE, RADIAL ARM, PHOTOGRAMMETRIC SET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an assortment of slotted template arms, arm clamps, bolts, nuts, washers, pins, studs, and tools which make up a slotted template set, and a chest to contain the various components. For purposes of clarity, the various items contained in the chest will be called a slotted template set throughout this specification.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GGG-H-33	- Hammers, Hand and Mallets, Surface Protective; Holders and Adapters, Inserted Hammer Face, and Faces, Hammer, Inserted.
GGG-P-471	- Pliers.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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GGG-W-641	- Wrench, Socket; (And Sockets, Handles and Attachments for Socket Wrenches; Hand).
GGG-S-291	- Shears, Metal Cutting, Bench and Hand.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-S-14197	- Stud, Photogrammetric, Slotted Templet.
MIL-P-14250	- Pin, Straight, Headed and Pin, Straight, Headless, Slotted Templet Stud.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13205E3600	- Slotted Template, Radial Arm, Photogrammetric: With Chest.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The slotted template set shall consist of the items shown on TA13205E3600, and the items listed in 3.7, and shall be as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the templates. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vender to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first-produced slotted template set). The contractor shall furnish one or more slotted template sets for examination and testing within the time frame specified (see 6.2), to prove that his production methods will produce slotted template sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Radial arm alinement. When tested as specified in 4.5.2.1, the perpendicular distance between the two marked positions of the orientation tip shall not exceed 0.01 inch, and the distance between the two marked circumferences of the same clamping hole shall not exceed 0.01 inch.

3.5 Arm clamps. The arm clamps shall fit the extended arms so that there is no arm movement in any direction except along the long axis of the slot.

3.6 Assembly. When tested as specified in 4.5.2.3, the slotted template radial arms shall assemble on the center bolt without sticking, binding, or stripping the threads. The radial arms of the assembled template shall not slip or turn on the center bolt.

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3.7 Additional components. The following components, in addition to those specified on the drawings, shall be furnished and placed in their proper places in the chest:

Item No.	Description	Identification	Quantity
1.	Hammer, hand	GGG-H-33, Type I, Class 3, Style D, Design a, 6 oz.	1
2.	Pin, headed	MIL-P-14250, Type I	1 gross
3.	Pin, headless	MIL-P-14250, Type II	500 (app. 1 lb)
4.	Pliers, common	GGG-P-471, Type II, Class 2, Style A, 6 inch nominal size.	1
5.	Shears, metal cutting	GGG-S-291	1
6.	Stud, photo-grammetric	MIL-S-14197	9 gross
7.	Wrench, socket	GGG-W-641, Type I, Class 3, 5/8 inch socket opening.	1

3.8 Workmanship. All parts of the slotted template shall be clean and free from pits, scale, scratches, burrs, and other harmful extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced slotted template set inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The individual components and the complete set shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The slotted template components shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection. Inspection shall be in two parts; inspection of components prior to assembly into sets, and after assembly into sets.

4.4.1 Component inspection. The nuts, bolts, washers, arm clamps, and each length of radial arm shall be considered separate units for the purpose of examination and tests.

4.4.1.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level I.

4.4.1.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined for the defects specified in 4.5.1. AQL shall be 0.40 percent defective.

4.4.1.3 Tests. Samples selected in accordance with 4.4.1.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.4.2 Slotted template set inspection.

4.4.2.1 Sampling. Sampling for examination of sets shall be in accordance with MIL-STD-105.

4.4.2.2 Examination. Samples selected in accordance with 4.4.2.1 shall be examined for the defects specified in 4.5.1. AQL shall be 1.0 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. The slotted template set shall be examined as specified herein for the following defects:

Component examination

101. Component not in accordance with the applicable drawings.
102. Treatment and painting of the chest not as specified.
103. Components not finished as shown on the drawings.
104. Workmanship not as specified.

Slotted template set examination

105. Components missing.
106. Components not placed in the chest as shown on the loading plans.

4.5.2 Tests.

4.5.2.1 Radial arm alinement. This test is to determine the accuracy to which the punched slot, the orientation tip, and the clamping holes are positioned on a common centerline. Fasten a sheet of drawing paper, drafting, or scribing medium, not less than 6 by 24 inches in size, to a drafting table top or other wooden working surface. Using either a hard-lead pencil, scriber or a ruling pen and a straightedge, draw a reference line lengthwise in the approximate center of the sheet. Starting 2 inches from one end, mark six points spaced 3/4 inch apart along the line. Drive headless pins into the table top, exactly on the reference line at the six marked points. Place six studs on the pins. Place a radial arm on the studs, along the reference line until the clamping holes are at their maximum distance from the studs. With the hard-lead pencil, scriber, or ruling pen and ink, trace the position of the orientation tip and the three clamping holes on the drafting medium. Remove the arm, turn it over, and replace the arm on the studs as stated above. Again trace the position of the orientation tip and the clamping holes on the paper. Measure the distance between the two traced positions of the orientation tip and the clamping holes. For testing arms shorter than 7-1/2 inches, the test shall be the same except the number of pins and studs required shall be as follows:

Arm length (Inches)	Number of pins and studs required
6	5
4-1/2	3
3-1/4	3
2	2

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When one arm fails to meet the requirements for alinement in 3.4, a second sample shall be tested. Evidence of one arm of the second sample failing to conform to the requirements of 3.4 shall constitute failure of this test.

4.5.2.2 Arm clamps. Not less than five extended arms shall be constructed using the 12- and 21-inch arms and two arm clamps for each extended arm. The arm clamps shall be examined for proper fit to the arms and for proper clamping action. The arm clamps shall be judged to fit the arms properly if the clamps allow movement of the arms along the long axis of the slot but no movement in any other direction. When any arm clamp fails to hold the arms properly, or if the clamps allow movement of the arms in any direction other than along the long axis of the slot, or when any arm clamp fails to hold the arms properly or is too tight to accept the two arms, retest shall be permitted. Evidence of any clamp during the retest failing to properly clamp the arm or evidence of any clamp not accepting two arms shall constitute failure of this test.

4.5.2.3 Assembly. Two templates shall be assembled, one with 8 arms (2 inches through 12 inches in length), and the other with 6 arms (13-1/2 inches through 21 inches). The arms shall be assembled on the center bolt in approximate 45 degree angles from 0 degree to 360 degrees. After the arms are assembled on the center bolt the hexagonal washer shall be placed on the center bolt. The double hexagonal wrench which prevents transmission of torque to the arm assembly is placed so that both the hexagonal bottom of the center bolt and the washer are held. The hexagonal nut shall be placed on the center bolt and tightened with the socket wrench while the double hexagonal wrench is used to hold the bolt and washer. The assembled template is then lifted and examined. Any slipping or turning of the arms on the center bolt or stripping of threads on either assembly shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified.	Each incorrect material, method, or container shall be considered one defect.

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108. Preservative not applied to surfaces as required by the referenced document for Level A.
109. Components not placed in the chest as provided.
110. Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Components of the slotted template set, except for the radial slotted arms, shall be preserved in accordance with MIL-P-116, Method IC-1. Components of like description shall be preserved together. Metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-7 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116. The components shall be placed in the chest, and the chest lid closed and secured.

5.1.2 Commercial. The components of the slotted template, contained in the chest as provided for, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A or Commercial, as specified (see 6.2).

5.2.1 Level A. The chest, containing all the components of the slotted template, shall be packed in a close-fitting box conforming to PPP-B-636, V3s, style optional. The box shall be closed and strapped in accordance with the appendix to the box specification.

5.2.2 Commercial. The chest, containing all components of the slotted template, shall not require packing. The chest lid shall be secured in a manner to prevent accidental opening or pilferage of the contents.

5.3 Marking. Chests and shipping containers shall be marked in accordance with MIL-STD-129 for Level A packaging, and MIL-STD-1188 for Commercial packaging.

6. NOTES

6.1 Intended use. The slotted template radial arm, photogrammetric set is intended for use in constructing radial-line templates from aerial photographs (overlapping vertical or trimetrogon), for establishing secondary control points, by radial triangulation, for laying controlled mosaics and other mapping and charting operations.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced template set, and number of sets required (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article (first-produced) model. Any changes or deviations of production slotted template sets from the approved first article (first-produced) model during production will be subject to the approval of the contracting officer. Approval of the first article (first-produced) model will not relieve the contractor of his obligation to furnish slotted template sets conforming to this specification.

6.4 Recyled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-11039E Slotted Template, Radial Arm, Photogrammetric Set

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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